

L12: Knowledge Check

1

Multiple choice 1 point

Recall that in the link-copy model, with probability p we connect the newly added node to a randomly chosen node, and with probability $q = 1 - p$ we "copy" an existing link. We covered the case $m = 1$, where each new node has one outgoing edge. Suppose that $m = 2$. How would you modify the differential equation that expresses the rate at which any node- j gains edges?

☐ $\frac{dx_j}{dt} = \frac{p}{t} + \frac{qx_j}{t}$

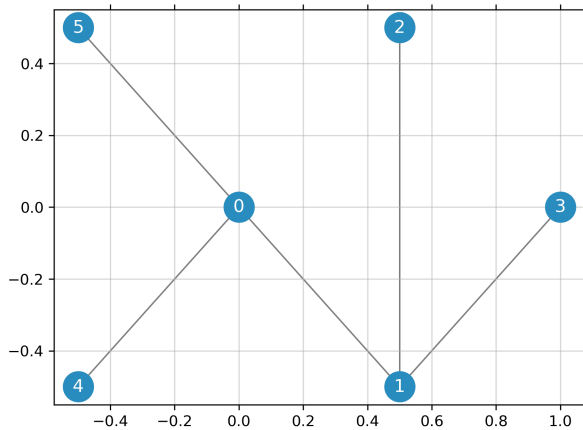
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This question relates to the following important point: *different network generation models (e.g., preferential attachment, link-copy, optimization-based) can generate the same network*. The following question gives you a network -- the nodes are added in the network in the order given by their labels (i.e., node-0 is added first, node-5 is added last). Each new node connects to one existing node. Which of the following statements is/are **TRUE or FALSE**?

(To apply the link-copy model on the undirected network, we can regard undirected networks as directed networks with edges in both directions, copy a directed edge, and then transform it into an undirected edge).



1. It is possible to generate this network with the optimization-based model we covered in this Lesson, with cost function $C_j = d_{i,j} + h_j$

2. It is possible to generate this network with the optimization-based model we covered in this Lesson, with cost function

$C_j = 0.1d_{i,j} + h_j$.

3. Suppose that the two candidate models are: linear preferential attachment, and the (undirected) link-copy model with $p = 1/2$. When adding node 4 in the network, the link-copy model assigns higher probability to edge (4, 0) than the linear preferential attachment model.

4. Suppose that the two candidate models are: linear preferential attachment, and the (undirected) link-copy model with $p = 1/2$. When adding node 5 in the network, the link-copy model assigns higher probability to edge (5, 0) than the linear preferential attachment model.

⋮ FALSE

⋮ TRUE

⋮ FALSE

⋮ TRUE

⋮ FALSE

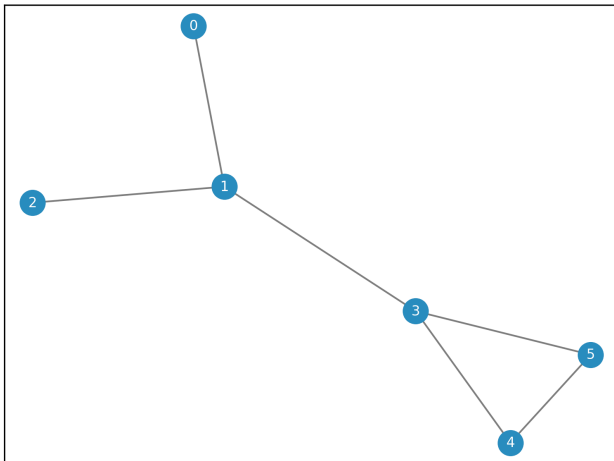
⋮ TRUE

⋮ FALSE

⋮ TRUE

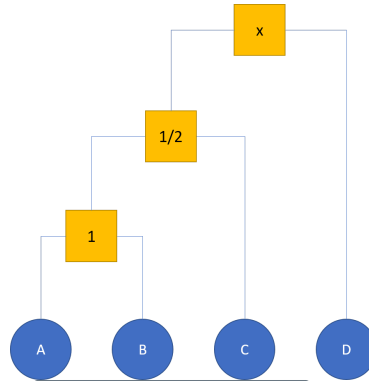
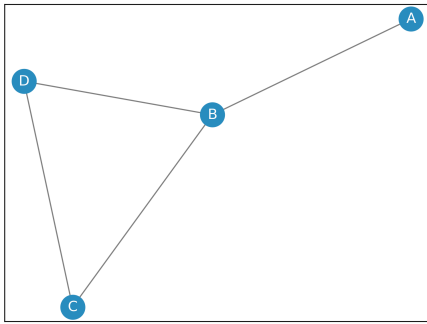
In the "Generating Networks with Degree Correlations" section of this Lesson, we described an algorithm to create degree correlations by rewiring.

If multi-edges are prohibited, which of the following statements is/are TRUE or FALSE for the following network?



1. We can apply that algorithm to make the network more assortative.
2. We can apply that algorithm to make the network more disassortative.
3. In the original network, the higher-degree nodes have lower average neighbor degree.
4. If the network is rewired to become maximally assortative, the higher-degree nodes will have higher average neighbor degree.

The following network was generated by the Hierarchical Random Graph model with the given dendrogram. Which of the following statements is/are **TRUE**?



1. The maximum-likelihood estimate of x is $\frac{2}{3}$.
2. With $x = \frac{2}{3}$, the likelihood of the model is $\frac{1}{27}$.
3. The given dendrogram is optimal -- we cannot increase the likelihood of the model with another dendrogram.
4. Swapping the position of nodes C and D in the dendrogram will not change the model's likelihood.

☐ FALSE

☐ TRUE

☐ FALSE

☐ TRUE

☐ FALSE

☐ TRUE

☐ FALSE

☐ TRUE

Which of the following is/are **TRUE or FALSE** about the Markov Chain Monte Carlo (MCMC) stochastic optimization method, and its application on finding the optimal dendrogram of the HRG model?

1. In each iteration of MCMC, comparing $L(D)$ with $L(D')$ is equivalent to comparing $L(D_r)$ with $L(D'_r)$, where D_r represents the subtree of D that is rooted at the parent of r .

2. If we generate a dendrogram D' from D using MCMC, and $L(D') \leq L(D)$, we may accept D' with probability $\frac{L(D')}{L(D)}$.

3. The main reason that we use MCMC is that it gives us a way to examine all possible dendrograms.

4. For two possible dendrograms D and D' , if D' is sampled by MCMC N times, then D is expected to be sampled by MCMC about $N \frac{L(D)}{L(D')}$ times.

⋮ FALSE

⋮ TRUE

⋮ FALSE

⋮ TRUE

⋮ FALSE

⋮ TRUE

⋮ FALSE

⋮ TRUE

Which of the following statements is/are true for the network models we covered in this lesson?

1. In the linear Preferential Attachment (PA) model a late node can never become the largest-degree hub -- but this is not true for the non-linear PA model with large α .

2. The Link-Copy model offers us a less flexible mechanism to generate networks with power-law distribution compared to the PA model.

3. We can create an assortative network by rewiring its edges as described in the "Generating Networks with Degree Correlations" section. The average neighbor node degree, as a function of node degree, will then be a monotonically increasing function.

4. The Hierarchical Graph Model allows us to capture various properties of real-world networks. However, its drawback is that it requires the estimation of a large number of parameters as well as a dendrogram.

⋮ FALSE

⋮ TRUE

⋮ FALSE

⋮ TRUE

⋮ FALSE

⋮ TRUE

⋮ FALSE

⋮ TRUE